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ON THE DENSITY OF THE PRIMES.

By ALDIN

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Aldin, of Lataki, whom the Society recognized as an Academician in the past year, sent the following communication:

ON THE DENSITY OF THE PRIMES.

The Society has done me the honour, in the past year, of recognizing me as an Academician. I think I cannot better show my thanks than by making early use of the leave it has thereby given me, and laying before the Society an investigation. The subject I have chosen is the density of the primes. Severin placed it first among his twenty-four questions; Crossley, nearly two hundred years ago, proved its approximate law; and many have studied it since. It may not, perhaps, be unworthy of such a purpose.

What Crossley proved is that the number of primes up to a large number x is nearly x divided by its logarithm; and that a closer approximation is the smooth counting formula, which adds up, piece by piece, the density $\frac{1}{\log t}$ at every place. How far the true number differs from the smooth counting formula he left to those who should come after. I shall show here that this difference is governed by the roots of a certain function, and that its size depends upon where those roots lie.

I. THE DENSITY FUNCTION

Crossley's proof makes use of an equation which joins all the primes with all the whole numbers:

$$\prod_{p} \frac{1}{1 - p^{-s}} = \sum_{n=1}^{\infty} \frac{1}{n^s} \tag{1}$$

The product on the left is taken over all primes p, and the sum on the right over all positive whole numbers n. The equation holds because every whole number is the product of primes in one way only. Crossley allowed s to take real values greater than one alone. I let s be a complex number, $s = \sigma + ti$; so long as its real part σ exceeds one, both sides converge and are equal as before. I denote it by $\zeta(s)$ and call it the density function, since the whole density of the primes is written in it.

Equation (1) gives $\zeta(s)$ only where $\sigma > 1$. To learn its form elsewhere we first write it otherwise. Let Γ be the function that extends the factorial to complex values, $\Gamma(n) = (n-1)!$. When $\sigma > 0$, from

$$\int_0^{\infty} e^{-nx} x^{s-1} dx = \frac{\Gamma(s)}{n^s} \tag{2}$$

we obtain, by summing over n when $\sigma > 1$,

$$\Gamma(s)\zeta(s) = \int_0^\infty \frac{x^{s-1}}{e^x - 1} dx \tag{3}$$

II. EXTENSION TO THE WHOLE PLANE

Consider now the integral $\int_\infty^\infty \frac{(-x)^{s-1}}{e^x-1} dx$, taken along this path: from positive infinity, close above the positive real axis, to the neighbourhood of the origin; once round the origin in the positive sense; and back, close below the positive axis, to positive infinity. The circuit is to be small, so as not to enclose the other zeros $\pm 2\pi i, \pm 4\pi i, ...$ of $e^x - 1$. By $(-x)^{s-1}$ is meant $e^{(s-1)\log(-x)}$, the logarithm of $-x$ being real when x is negative. This integral has a definite value for every complex s, and may everywhere be expanded term by term. When $\sigma > 1$, the path may be drawn down upon the axis: the part round the small circuit then vanishes, and the parts above and below the axis together give $-2i\sin(\pi s)$ times the integral in (3); and we find

$$2\sin(\pi s)\Gamma(s)\zeta(s) = i\int_\infty^\infty \frac{(-x)^{s-1}}{e^x - 1} dx \tag{4}$$

The integral on the right, once the radius of the circuit is fixed at less than 2π, has a definite value for every s and is everywhere smooth. The factor $\sin(\pi s)\Gamma(s)$ on the left, however, has its zeros, and the division must be made with care. Since $\Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin(\pi s)}$, equation (4) may be written $\zeta(s) = \frac{i\Gamma(1-s)}{2\pi}\int_\infty^\infty \frac{(-x)^{s-1}}{e^x-1} dx$, so that $\zeta(s)$ can become infinite only where $\Gamma(1-s)$ does, at $s = 1, 2, 3, ...$ At $s = 2, 3, ...$, however, $\zeta(s)$ is already given finite by (1), and there the integral vanishes; only at $s = 1$ does it truly become infinite, as $\frac{1}{s-1}$ does. Thus $\zeta(s)$ is extended over the whole complex plane.

III. THE SYMMETRY

The function $\zeta(s)$ has besides a symmetry, which joins s with $1 - s$. Let

$$\psi(x) = \sum_{n=1}^\infty e^{-n^2\pi x}$$

The tide-reckoners of the Tower of All Things, in parting a record into its constituent tides, make frequent use of an equation which here becomes, for positive x,

$$2\psi(x) + 1 = x^{-\frac{1}{2}}\left(2\psi\left(\frac{1}{x}\right) + 1\right) \tag{5}$$

Putting $\frac{s}{2}$ for s and $n^2\pi x$ for x in (2), we have $\int_0^\infty e^{-n^2\pi x} x^{\frac{s}{2}-1} dx = \pi^{-\frac{s}{2}} n^{-s}\Gamma\left(\frac{s}{2}\right)$; and summing over n when $\sigma > 1$, $\pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s) = \int_0^\infty \psi(x) x^{\frac{s}{2}-1} dx$. Divide this integral at $x = 1$, and carry the first part beyond one by means of (5); there results

$$\pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s) = \frac{1}{s(s-1)} + \int_1^\infty \psi(x)\left(x^{\frac{s}{2}-1} + x^{\frac{1-s}{2}-1}\right) dx \tag{6}$$

The right side is unchanged when $1 - s$ is put for s; therefore

$$\pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s) = \pi^{-\frac{1-s}{2}}\Gamma\left(\frac{1-s}{2}\right)\zeta(1-s) \tag{7}$$

Equations (6) and (7) are first derived where $\sigma > 1$; since the right side of (6) has a meaning for every s other than 0 and 1, both hold everywhere. By (7), when s is $-2, -4, -6, ...$, the factor $\Gamma(\frac{s}{2})$ on the left becomes infinite while the right side is finite; therefore $\zeta(s)$ vanishes at these places. These I call its trivial roots, and the others its non-trivial roots.

IV. THE ROOTS

Put $s = \frac{1}{2} + ti$, and let

$$\xi(t) = \frac{1}{2}s(s-1)\pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s) \tag{8}$$

By (7), $\xi(t)$ is unchanged when $-t$ is put for t; by (6), it has a definite value everywhere in the plane and nowhere becomes infinite, and it may be written as a series in powers of t^2 which converges very rapidly; and for real t it takes real values. The non-trivial roots of $\zeta(s)$ are exactly the roots of $\xi(t)$. The imaginary part of t at each of these roots lies between $-\frac{1}{2}$ and $\frac{1}{2}$; in other words, the non-trivial roots all lie in the band between the lines $\sigma = 0$ and $\sigma = 1$, and lie symmetrically on either side of the line $\sigma = \frac{1}{2}$. There is no non-trivial root outside the band: where $\sigma > 1$ the logarithm of the product (1) converges absolutely, and ζ does not vanish; where $\sigma < 0$ there is none, by the symmetry (7). Whether there is a root upon the two edges of the band requires another argument. I believe that there being no root upon the edges is the same thing as Crossley's approximate law, that is, that the ratio of the number of primes to x divided by $\log x$ tends to one; but this must be proved separately, and I shall not go into it here.

The number of non-trivial roots, counted according to their multiplicity, whose t has its real part between 0 and T is approximately

$$\frac{T}{2\pi}\log\left(\frac{T}{2\pi}\right) - \frac{T}{2\pi} \tag{9}$$

and the error is of no greater order than $\log T$. The number may be found by carrying ξ once round the boundary of a rectangle and observing how far its angle has turned: the principal term comes from the angle of $\pi^{-\frac{s}{2}}\Gamma(\frac{s}{2})$ along the line, and on the other sides the angle of ζ changes by no more than of the order of $\log T$. Where a side passes through a root, the side is moved a little.

I have reckoned the roots with T up to 30. I first prepared from (6) the series for ξ, with bounds for its remainders, and then asked the star-reckoning arrays of the Tower of All Things to unfold the tables by differences. Since $\xi(t)$ is real for real t, every change of its sign encloses at least one real root of odd multiplicity, lying upon the line $\sigma = \frac{1}{2}$; a root of even multiplicity, however, makes no change of sign, so that the changes of sign alone do not suffice. Between $t = 0$ and $t = 30$ I found three changes of sign, between 14.1 and 14.2, between 21.0 and 21.1, and between 25.0 and 25.1; at both ends of each the values far exceed the bound of the remainder, so that the signs cannot be wrong. Then, by the turning of the angle described above, I carried ξ round the rectangle in which the real part of t runs from 0 to 30 and its imaginary part from -1 to 1, taking a point at every twentieth part along the

sides; the value of ξ at every point lies far from zero, and between neighbouring points its angle turns by less than a quarter of a circuit, so that no whole turn can be missed. The angle turned three times in all: this stretch contains three roots, counted by multiplicity. The three changes of sign therefore each enclose a single root, and there is no other root, on the line or off it. Their places are given in Table I. Formula (9) gives nearly three at $T = 30$, in agreement.

TABLE I. The non-trivial roots with T up to 30

Order	t	*The root* s
First	14.13	1/2 + 14.13 i
Second	21.02	1/2 + 21.02 i
Third	25.01	1/2 + 25.01 i

It is very probable that all the non-trivial roots lie upon the line $\sigma = \frac{1}{2}$. One would of course wish for a strict proof of this. I have made some attempts without success, and have put the search aside for the present, since what follows does not require it.

V. THE NUMBER OF PRIMES

Let $F(x)$ be the number of primes less than x; when x is itself a prime, let $F(x)$ be the mean of its values just before and just after the step. Let further

$$f(x) = F(x) + \frac{1}{2}F\left(x^{\frac{1}{2}}\right) + \frac{1}{3}F\left(x^{\frac{1}{3}}\right) + \ldots \tag{10}$$

so that every prime is counted as one, every square of a prime as one half, every cube as one third, and so on. Taking logarithms of both sides of (1) we find

$$\log \zeta(s) = \sum_p \sum_{n=1}^{\infty} \frac{1}{n} p^{-ns} = s \int_1^{\infty} f(x) x^{-s-1} dx \tag{11}$$

This equation gives $\log \zeta$ from f; it may also be turned about, to give f from $\log \zeta$:

$$f(x) = \frac{1}{2\pi i} \int_{a-i\infty}^{a+i\infty} \log \zeta(s) x^s \frac{ds}{s} \tag{12}$$

the integral being taken along the line $\sigma = a$, with a and x greater than one, between the limits $a \pm iT$, T being then made to grow without end. Here $\log \zeta$ is defined where $\sigma > 1$ by the series in (11), and is real upon the positive real axis. At the powers of primes the integral gives the mean of the values just before and just after the step, in agreement with the convention of (10). When the tide-reckoners put a record together again from its constituent tides, they use such a pair of reckonings as this; only here it is the primes that are put together.

Now $\xi(t)$ may be written as $\xi(0)$ multiplied by the product of all the factors $\left(1 - \frac{t^2}{\alpha^2}\right)$, where α runs over the roots of ξ, α and $-\alpha$ being taken once only, and repeated roots according to their multiplicity. Two things are needed for this: by (6), $\xi(t)$ grows no faster than of the order of $e^{|t| \log |t|}$, so that the product converges; and since ξ is even, and grows so slowly, there can be no exponential factor besides the product. Accordingly $\log \zeta(s)$ falls into several parts: one arising from the infinity

at $s = 1$, one from Γ, and one from each of the non-trivial roots. These are put one by one into (12). The integral is first cut off at $a \pm iT$, with T taken between the heights of two roots, and the path pushed to the left; each root then gives a term, $s = 1$ gives $L(x)$, the part from Γ gives the last integral, and the constant gives $-\log 2$; and it is shown that the integrals along the upper and lower sides of the moved path vanish as T grows without end. We obtain

$$f(x) = L(x) - \sum_{\rho} L(x^{\rho}) - \log 2 + \int_{x}^{\infty} \frac{dt}{t(t^2-1)\log t} \tag{13}$$

Here $L(x)$ is Crossley's smooth counting formula,

$$L(x) = \int_{0}^{x} \frac{dt}{\log t}$$

the integral being taken, where it passes $t = 1$, as the limit of equal parts on either side. Crossley began his integral at 2; the two forms differ only by a constant, about 1.045, which may be neglected in the comparisons below. And ρ runs over the non-trivial roots, counted according to their multiplicity. $L(x^{\rho})$ is to be understood as $\int_{-\infty}^{\rho \log x} \frac{e^u}{u} du$, the path coming round from the left to $\rho \log x$, and not by first reckoning x^{ρ} and then taking its logarithm; the terms from the upper and lower halves of the plane are conjugate. The terms are taken up to the roots whose imaginary parts do not exceed T in magnitude, and T is then made to grow without end. Lastly, inverting (10), we have

$$F(x) = \sum_{m=1}^{\infty} \frac{\mu(m)}{m} f\left(x^{\frac{1}{m}}\right) \tag{14}$$

where $\mu(1)$ is one, $\mu(m)$ is zero when m has a square factor, and otherwise is one or minus one according as the number of prime factors of m is even or odd. For a given x the terms with m greater than $\frac{\log x}{\log 2}$ are all nothing, so that (14) has in truth only finitely many terms. Equations (13) and (14) together are an exact formula for the number of primes; I call them the density formula.

VI. THE ERROR

If the smooth principal term of the density formula is inverted by (14), there results

$$A(x) = \sum_{m=1}^{\infty} \frac{\mu(m)}{m} L\left(x^{\frac{1}{m}}\right) \tag{15}$$

$$= L(x) - \frac{1}{2} L\left(x^{\frac{1}{2}}\right) - \frac{1}{3} L\left(x^{\frac{1}{3}}\right) - \frac{1}{5} L\left(x^{\frac{1}{5}}\right) + \frac{1}{6} L\left(x^{\frac{1}{6}}\right) - \ldots$$

which I call the estimate of the density formula. The series (15) converges slowly, and for reckoning I use instead this equivalent form, which converges fast:

$$A(x) = 1 + \sum_{k=1}^{\infty} \frac{(\log x)^k}{k \cdot k! \cdot \zeta(k+1)}$$

When x is large, the estimate differs from the smooth counting formula only by $\frac{1}{2}L(\sqrt{x})$ and a few further terms, together of the order of $\frac{\sqrt{x}}{\log x}$.

The difference between $F(x)$ and the estimate comes, apart from a few small terms inverted from $-\log 2$ and from the last integral, which within the range of the table together amount to less than one, wholly from the non-trivial roots. Each root $\rho = \beta + \gamma i$ contributes a term whose envelope is about $\frac{x^\beta}{|\rho| \log x}$, and whose phase turns with $\log x$ in the period $2\frac{\pi}{\gamma}$; and the terms of different roots partly cancel one another. The roots go in groups: ρ and $1 - \overline{\rho}$ lie symmetrically on either side of $\sigma = \frac{1}{2}$, and the one of a group that lies to the right has real part $\frac{1}{2} + |\beta - \frac{1}{2}|$, which fixes how fast the swing of that group may grow. Thus the farther a group of roots lies from the line $\sigma = \frac{1}{2}$, the greater the swing it may give; how far the true number departs from the smooth counting formula depends wholly upon where the roots lie.

If all the non-trivial roots lie upon $\sigma = \frac{1}{2}$, the envelope of each term is $\frac{\sqrt{x}}{|\rho| \log x}$ only. To add them up it is convenient first to count otherwise, reckoning each power p^k of a prime as $\log p$, so long as p^k is less than x, and taking the mean at each step as before; call this count $H(x)$. Its exact formula is derived as (13) is, but is cleaner: $H(x) = x - \sum_\rho \frac{x^\rho}{\rho} - \log 2\pi - \frac{1}{2}\log(1 - x^{-2})$. If the roots are cut off where their imaginary parts exceed T in magnitude, the part cut off is of no greater order than $x\frac{(\log xT)^2}{T}$, and by (9) the part kept is of no greater order than $\sqrt{x}(\log T)^2$. Taking T comparable with $\sqrt{x}$, both parts are of no greater order than $\sqrt{x}(\log x)^2$; and in passing back from H to the number of primes one logarithm is removed. Hence:

When x is large enough, the true number of primes up to x differs from the estimate of the density formula by no more than a constant multiple of $\sqrt{x}\log x$; and from the smooth counting formula by an amount of the same order.

I conjecture that this is so. But I can derive it only on the supposition that all the roots lie upon the line, and that supposition I cannot prove. The converse holds also: if for all large x the error is no more than a constant multiple of $\sqrt{x}\log x$, then $H(x) - x$ is of no greater order than $\sqrt{x}(\log x)^2$, so that $-\frac{\zeta'(s)}{\zeta(s)} - \frac{s}{s-1} = s\int_1^\infty (H(x) - x)x^{-s-1}dx$ has a definite value everywhere where $\sigma > \frac{1}{2}$, and ζ has no root there; and by the symmetry the non-trivial roots all lie upon the line. If a single root lies off the line, then no constant can hold the error for all large x. The two matters are therefore one: to ask how great the error is, is to ask where the roots lie. The estimates above I have given only in outline, not in full.

VII. COMPARISON WITH THE TABLES OF PRIMES

The star-reckoning arrays of the Tower of All Things have sieved out the primes below ten million. Table II compares the counts of that table with the smooth counting formula and with the estimate of the density formula.

Table II. The number of primes below ten million, with the two formulae (the primes from the sieve tables of the arrays of the Tower of All Things)

x	*True number F(x)*	*Smooth count L(x)*	$L - F$	*Estimate A(x)*	$A - F$	$\sqrt{x}$
1,000,000	78,498	78,628	130	78,527	29	1,000
2,000,000	148,933	149,055	122	148,924	−9	1,414
3,000,000	216,816	216,971	155	216,816	0	1,732
4,000,000	283,146	283,352	206	283,179	33	2,000
5,000,000	348,513	348,638	125	348,449	−64	2,236
6,000,000	412,849	413,077	228	412,873	24	2,449
7,000,000	476,648	476,827	179	476,610	−38	2,646
8,000,000	539,777	540,000	223	539,771	−6	2,828
9,000,000	602,489	602,676	187	602,436	−53	3,000
10,000,000	664,579	664,918	339	664,667	88	3,162

The values of L and A are reckoned to sufficient places and then given to the nearest whole number; the columns of differences are taken between these whole numbers, so that the nothing at three million means only that the two are equal when so rounded.

At all ten points of the table the smooth counting formula is too great, by from a hundred and twenty to three hundred and forty, which is less than a seventh of $\sqrt{x}$. The excess is comparable with the term $\frac{1}{2}L(\sqrt{x})$, the ratio of the two lying between two thirds and one and a half, and the part by which they differ rises and falls with the roots. The estimate differs from the true number by less than ninety, sometimes above and sometimes below. That the differences are now positive and now negative is the work of the swinging terms of the roots.

Severin, looking at the primes below a hundred thousand, judged that they thin out "on the whole, evenly", while their small rises and falls "seem to have no law". It may now be said that the even part is the smooth counting formula, and that the small rises and falls have their law also: they are the swings given by the several roots of the density function, laid one upon another. Fig. 1 shows the number of primes below fifty, the value at each step being the mean, together with the estimate, and with the result of adding the first three roots, the term $L(x^\rho)$ of each being likewise inverted by (14) over every m, together with its conjugate root, and the places of the roots being used unrounded. When the three roots are added the curve begins to bend at the corners of the steps, but it is still far from tracing them; to trace the steps many roots must be added.

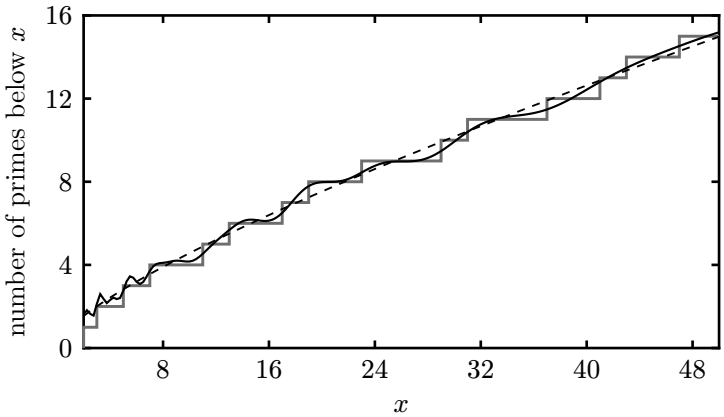


FIG. 1. The number of primes below fifty (steps), the estimate of the density formula (broken line), and the same with the first three roots (solid line)

Summed in the manner laid down above, the more roots are added, the more nearly the curve keeps to the steps away from the steps themselves, and to the mean at each step; but with finitely many roots the error need not lessen steadily as roots are added, and near the steps the curve may overshoot. To reckon more roots, and to check the tables of primes further on, is beyond the strength of one or two men. I hope that the Society may ask the star-reckoning arrays to carry the reckoning of the roots onward, and see whether they all lie upon the line. If one of them lies off it, my conjecture is false; if all that are reckoned lie upon it, that is still no proof.